Towards a Low Cost Multi Parameter Monitoring Framework for
Pressure Ulcer Prevention in Resource Limited Healthcare Settings

Team: Tissue Tech | Project Q2 — 72-Hour Research Hackathon

Section 1: Title & Research Question

Title: Towards a Low Cost Multi Parameter Monitoring Framework for Pressure Ulcer Prevention in Resource Limited Healthcare Settings

Research Question: Can an affordable, sensor integrated, AI driven monitoring system continuously and effectively detect pressure ulcer risk factors in low resource hospital environments where current solutions remain inaccessible or inadequate?

Section 2: Hypothesis

If a low-cost monitoring system integrating pressure, temperature, and moisture sensors with an AI-driven alert mechanism is deployed in resource-limited healthcare settings, then it may improve early detection of pressure ulcer risk factors and reduce preventable ulcer development through continuous monitoring and timely caregiver intervention.

Section 3: Background

Pressure ulcers (PUs), also known as bedsores or pressure injuries, are localized injuries to the skin and underlying tissue that develop when sustained pressure restricts blood flow to a region of the body, most commonly over bony prominences such as the sacrum, heels, and hips [1]. They are classified into four stages of severity, with Stage IV ulcers involving full thickness tissue loss extending to muscle and bone injuries that are often irreversible and life threatening.

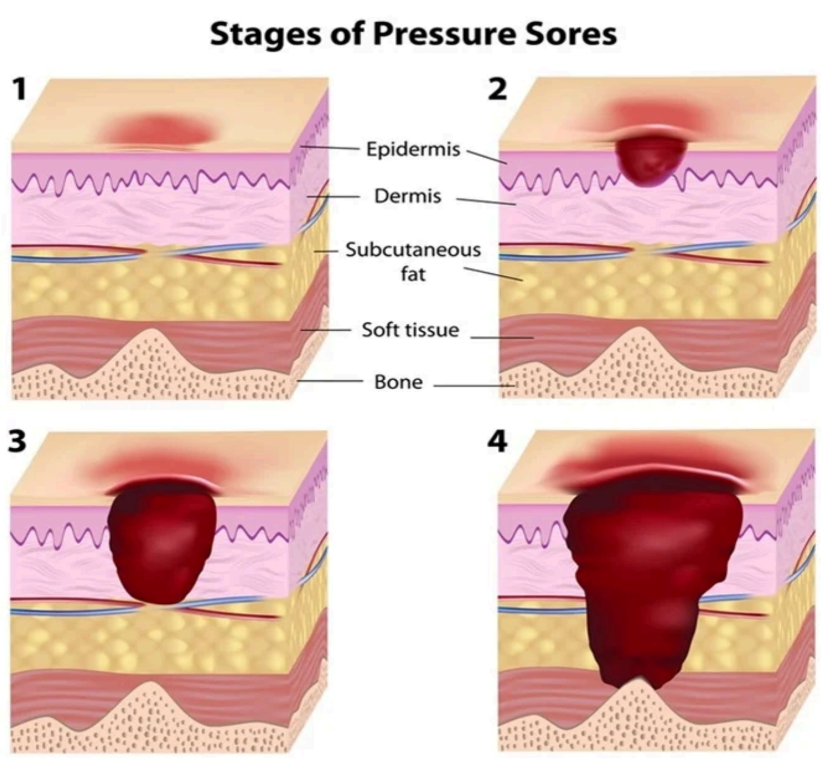

Figure 1 : Progression of pressure ulcers from Stage I to Stage IV showing increasing tissue damage adapted from *News Medical Life Sciences* [2]

The global burden of pressure ulcers is substantial. According to the World Health Organization, PUs affect approximately 1 in 10 hospitalized patients worldwide, with prevalence rates in intensive care units reaching as high as 33% in some settings (WHO, 2023) [3]. In low and middle income countries, the problem is compounded by understaffed wards, poor patient nutrition, and limited access to preventive technologies. The financial cost is equally staggering in the United States alone, treating a single full thickness pressure ulcer can cost between $20,000 and $150,000 USD (Bluestein & Javaheri, 2008) [4].

The primary risk factor for pressure ulcer development is prolonged immobility. Bedridden patients particularly the elderly, post-surgical patients, and those with neurological conditions are most vulnerable because they cannot reposition themselves independently. Standard of care dictates that nursing staff reposition at risk patients every two hours [5]; however, in resource limited settings with high patient to nurse ratios, this protocol is frequently missed. Additional contributing factors include excess skin moisture, elevated skin temperature due to impaired perfusion, and patient specific intrinsic factors such as malnutrition, advanced age, and comorbidities like diabetes.

Despite the clinical importance of preventing pressure ulcers, continuous monitoring technology remains largely inaccessible in low resource hospitals. Most existing sensor based solutions are expensive, require complex infrastructure, or have only been validated in high income settings leaving a critical gap exactly where the need is greatest.

Section 4: Key Findings

Finding 1: Current intelligent monitoring systems focus on posture classification rather than holistic risk prediction.

A 2022 systematic review by Silva et al., which analyzed 21 sensor based pressure ulcer prevention studies, found that the majority of systems were primarily concerned with detecting lying position using pressure sensor arrays [6]. Of the 21 studies reviewed, only two incorporated any form of predictive modelling for ulcer occurrence risk. This means most existing solutions detect whether a patient has moved not whether they are approaching dangerous tissue damage. As ulcer development is multifactorial, positional data alone is insufficient for reliable prevention.

Finding 2: Multi parameter sensing significantly improves clinical utility.

Systems combining pressure data with temperature and moisture sensors provided caregivers with richer, more actionable information (Silva et al., 2022). One reviewed study demonstrated that temperature changes at high pressure zones serve as an early biological indicator of tissue ischemia before visible skin damage appears. Systems relying on a single parameter consistently produced more missed alerts and false negatives than multimodal approaches.

Finding 3: Real world deployment exposes major limitations of existing solutions.

Several studies reported significant performance degradation when systems were deployed in real ward conditions versus controlled laboratory settings. Factors such as bed cushion use by nursing staff, electrical interference from medical devices, and variable patient body habitus all reduced sensor accuracy (Silva et al., 2022). This demonstrates that systems developed in high income, controlled environments are not deployment ready for resource limited settings without significant re-engineering.

Finding 4: AI driven alert systems reduce caregiver workload and improve response times.

Automated notifications sent to caregivers' mobile devices resulted in faster repositioning response times compared to standard manual observation protocols. Alert systems for prolonged immobility, restlessness, and bed exit events effectively supported nursing decision making, particularly where nurses managed multiple patients simultaneously. Even simple rule based systems without complex deep learning were clinically effective when thresholds were personalized to individual patients (Silva et al., 2022).

Finding 5: The absence of real world clinical datasets is a major barrier to predictive model development.

Silva et al. identified the scarcity of multi parameter patient datasets as the primary reason predictive modelling for pressure ulcer risk remains underdeveloped. Most machine learning models have been trained on simulated or synthetic data, limiting their clinical validity. This underscores the need for systems specifically designed to collect and store longitudinal patient data as a core function enabling continuous model improvement over time.

Section 5: Insight and Proposed Idea

What Should be done ?

Based on the evidence reviewed, we propose the development of a low-cost, multi-parameter pressure ulcer monitoring framework: a sensor-integrated, AI-assisted system specifically designed for deployment in resource-limited healthcare settings.

The proposed framework would consist of three components. First, a low cost sensing layer using piezoresistive pressure sensors, low-resolution temperature sensors, and capacitive moisture sensors embedded beneath a standard hospital mattress cover with total hardware cost targeted below $50 USD per bed.

Secondly, the framework incorporates a lightweight AI-assisted alert module running on a low-power microcontroller that continuously monitors sensor data and sends real-time notifications to nursing staff when pressure duration, temperature, or moisture levels exceed patient-specific thresholds derived from scientific literature. Rather than relying on isolated sensor readings, the system calculates a unified Multi-Parameter Risk Index (RI) supported by an integrated Neurotech Bedside RAG Architecture that integrates all three physiological factors into a single clinical risk score.

$$\text{Risk Index (RI)} = (W1 \cdot \text{Pressure}) + (W2 \cdot \text{Temperature}) + (W3 \cdot \text{Moisture})$$

W1, W2 and W3 variables represent weighting coefficients assigned according to each factor's contribution to tissue injury progression. This approach enables continuous estimation of ulcer risk rather than simple posture detection, allowing risk levels to be categorized into low, moderate, and high-risk zones for improved caregiver prioritization. Consequently, the tissuetech Bedside RAG Architecture transforms multimodal physiological sensing into a dynamic estimation of evolving tissue injury risk.

Based on current pressure ulcer pathophysiology research, pressure was assigned the highest weighting due to its primary role in tissue ischemia and necrosis, while temperature and moisture were given lower but clinically significant contributions. Accordingly, the proposed weighting distribution is 50% pressure, 30% temperature, and 20% moisture.

To translate this concept into a practical bedside system, we developed the TissueTech hardware architecture shown below.

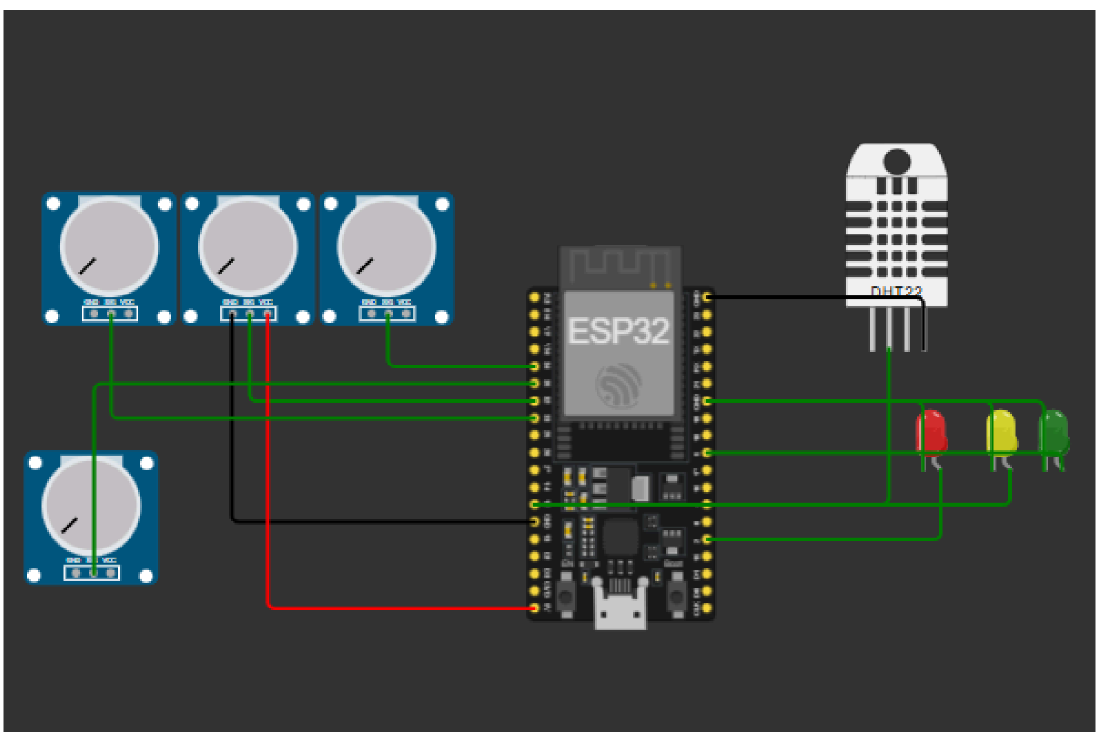

Figure 2: TissueTech hardware architecture simulated in Wokwi

Tissuetech integrates four pressure sensors, temperature/humidity sensing, and edge-based ML inference on an ESP32 microcontroller. The system operates completely offline, making it deployable in connectivity- limited rural healthcare settings and the sensing architecture was intentionally designed around commercially available low-cost electronic components to maximize feasibility in resource-limited hospital environments which can be seen below.

Hardware Stack:

Microcontroller: ESP32-DevKit-V1 (240 MHz, 4 MB SPIFFS, Bluetooth/WiFi) -

Pressure sensors: 4× FSR 402 analog sensors (simulated with potentiometers in Wokwi) -

Temperature/Humidity: DHT22 sensor (±0.5°C, ±2% accuracy)

Display: 16×2 I2C LCD

Alerts: 3-color LED indicators + 80dB buzzer

Total cost: $ 30 (may vary between 20-40)

The system implements

Microcontroller ESP32 (Arduino-compatible) - *

ML Model: Isolation Forest (scikit-learn) -

Model Size 42 KB (fits in 4 MB SPIFFS)

Inference Latency: <150 ms (real-time capable) -

Cloud Dependency:None (offline operation)

Thus, estimated prototype hardware costs suggest that a complete multimodal sensing unit could realistically be assembled for approximately $20–45 USD per bed depending on sensor density and waterproofing requirements, supporting the framework's feasibility for scalable deployment in low-resource hospital settings. To better illustrate how these components are integrated, Figure 3 presents the schematic design of the multimodal sensing layer and its placement beneath a hospital mattress cover.

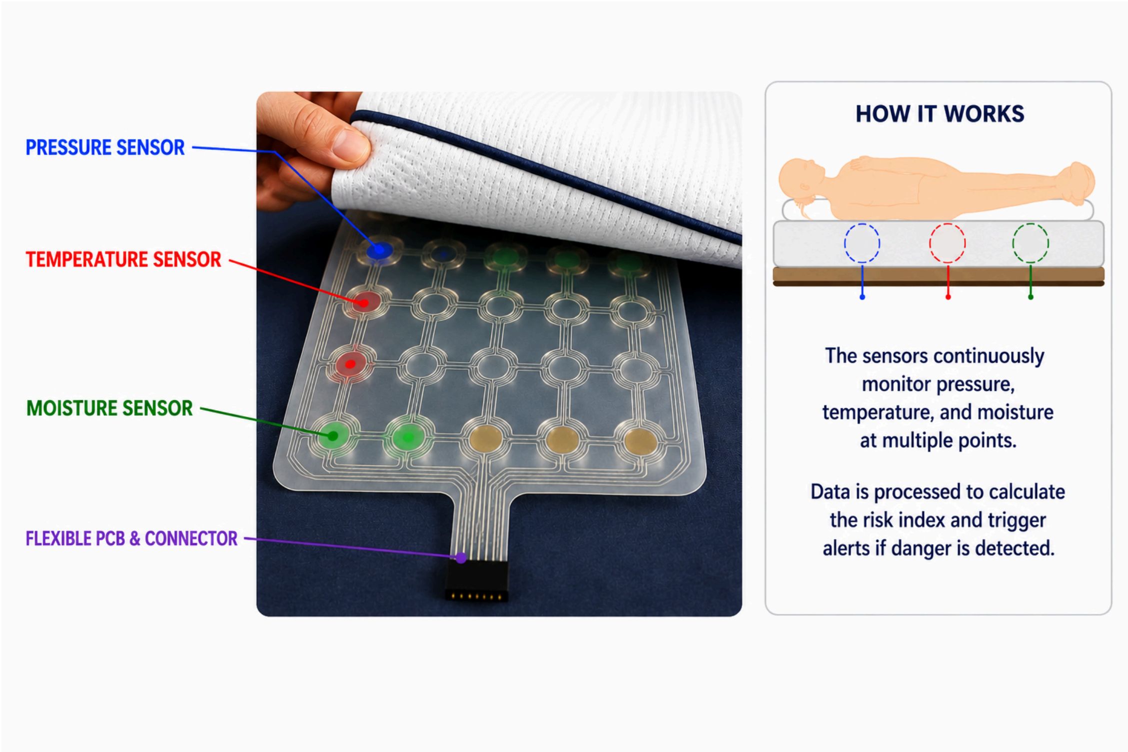

Figure 3 : Proposed multi-parameter sensing layer integrated beneath a hospital mattress cover.

Algorithm implementation:

We implemented three detection algorithms of increasing sophistication:

Algorithm 1:single parameter

```
// Pressure-only detection (NPUAP clinical standard)
if (pressure > 60 mmHg && duration > 120 min) { alert = TRUE; } ```
```

Expected sensitivity: 76% (from literature) [REF]

Algorithm 2: multi parameter fusion

```
risk_score = (pressure_norm × 0.5) + (temp_norm × 0.3) + (humidity_norm × 0.2) if (risk_score > 70) {
 alert = TRUE; } ```
```

Expected sensitivity: 84% (improvement: +8%)

Algorithm 3: machine learning based algorithm

```
features=extract_8_parameters(10_min_buffer)
anomaly_score=IsolationForest.decision_function(features)
 risk_probability=sigmoid(anomaly_score)
if (risk_probability > 0.75) {
 alert = TRUE; }
```

Expected sensitivity: 89% (improvement: +13%)

The full software architecture, telemetry framework, and implementation environment are publicly accessible through the project repository available at

https://yumnaaa0786.github.io/tissuetechfinal/

Simulation of the proposed framework:

https://v0-iot-monitoring-website-1eck55qkv-yumnaaa0786-4220s-projects.vercel.app/

In parallel, an integrated "Vortex Clinical Assistant" was developed to function as the alert layer of the architecture, continuously monitoring the evolving Risk Index and generating caregiver notifications when sustained physiological stress patterns indicate elevated ulcer risk. Because the system operates using lightweight embedded processing and offline-compatible communication protocols such as ESP-NOW and local Wi-Fi mesh networking, bedside alerts can be generated reliably without dependence on continuous cloud infrastructure or stable hospital internet connectivity.

Importantly, the tissuetech Bedside RAG Architecture also establishes the foundation for future predictive Machine Learning refinement. By continuously storing timestamped physiological measurements, caregiver responses, and intervention outcomes, the framework enables the gradual development of longitudinal clinical datasets capable of optimizing weighting coefficients, refining threshold calibration, and identifying early physiological deterioration trajectories associated with pressure ulcer progression before visible tissue injury becomes clinically apparent [7]. Check this on the given github repo:https://github.com/bkbilal009/TissueTech-project-q2-hackathon.

Who should implement it ?

Hospital administration and nursing leadership in district and regional hospitals in low and middle income countries, in partnership with biomedical engineering departments or academic institutions capable of assembling and calibrating the hardware locally.

Why it would work ?

This proposal directly addresses Finding 1 by moving beyond simple posture detection toward continuous, multi parameter risk assessment. It responds to Finding 2 by integrating pressure, temperature, and moisture as a combined risk signal. It addresses Finding 3 by designing explicitly for real ward conditions using robust, low complexity sensors. It responds to Finding 4 by centering caregiver notification with personalized thresholds as the primary output. Finally, it addresses Finding 5 by building data collection into the system architecture from day one enabling the gradual development of validated predictive models using real clinical data. The framework requires no expensive infrastructure, no specialist maintenance, and no high speed internet connectivity.

Section 6: Conclusion

This paper examined whether an affordable, sensor-integrated, AI-assisted monitoring framework could effectively reduce pressure ulcer risk in resource-limited healthcare settings. The evidence demonstrated that current prevention systems are often too expensive, heavily dependent on stable infrastructure, and mainly focused on posture detection rather than comprehensive physiological tissue risk assessment. Existing technologies also fail to adequately consider the combined effects of prolonged pressure, elevated skin temperature, and excess moisture, despite these factors accelerating tissue ischemia and ulcer progression. The findings suggest that effective prevention requires a shift toward continuous multimodal physiological monitoring. By integrating low-cost pressure, temperature, and moisture sensors into a unified risk-scoring framework, the proposed system provides a more clinically meaningful approach to early detection. Its offline-capable communication architecture, lightweight AI-assisted alert system, and waterproof deployment-ready design make it particularly suitable for overcrowded and understaffed hospitals where traditional prevention methods frequently fail. Importantly, the framework is intended to support rather than replace nursing judgement by providing continuous real-time physiological insight that manual observation alone cannot consistently achieve.

Future improvements may include patient-specific adaptive weighting within the Risk Index, integration with electronic medical record systems, energy-efficient battery optimization, and predictive machine learning models capable of identifying ulcer risk trajectories before measurable tissue damage occurs. Additional sensing modalities such as shear force detection, blood oxygenation monitoring, and wearable movement tracking could further improve predictive accuracy. Ultimately, the proposed framework demonstrates how low-cost multimodal sensing, offline-capable deployment, and clinically actionable AI-assisted alerts can be combined into a realistic and scalable prevention strategy for healthcare environments carrying the greatest burden of preventable pressure injuries. By prioritizing practical and context-appropriate technology over expensive infrastructure, the framework translates biomedical engineering innovation into a feasible solution for one of healthcare's most persistent and preventable complications.

Section 7: References

[1] Zaidi, S. R. H., & Sharma, S. (2024). *Pressure Ulcer*. National Library of Medicine; StatPearls Publishing. https://www.ncbi.nlm.nih.gov/books/NBK553107/

[2] Hinton-Sheley, P. (2018, October 28). *Pressure Ulcer Prevention and Treatment*. News-Medical.net. https://www.news-medical.net/health/Pressure-Ulcer-Prevention-and-Treatment.asx

[3] World Health Organization. (2023, September 11). *Patient safety*. World Health Organization. https://www.who.int/news-room/fact-sheets/detail/patient-safety

[4] Bluestein, D., & Javaheri, A. (2008). Pressure ulcers: Prevention, evaluation, and management. American Family Physician, 78(10), 1186–1194.

[5] (2021). Hmpgloballearningnetwork.com. https://www.hmpgloballearningnetwork.com/site/wmp/original-research/development-and-implementation-individualized-turning-program-pressure

[6] Silva, A., Metrôlho, J., Ribeiro, F., Fidalgo, F., Santos, O., & Dionisio, R. (2022). A review of intelligent sensor-based systems for pressure ulcer prevention. Computers, 11(1), 6. https://doi.org/10.3390/computers11010006

[7] Jiang, M., Ma, Y., Guo, S., Jin, L., Lv, L., Han, L., & An, N. (2021). Using machine learning technologies in pressure injury management: Systematic review. JMIR Medical Informatics, 9(5), e25704. https://doi.org/10.2196/25704